PHASE 4 EDUCATIONAL PACKAGE JCE TECHNOLOGY REPORT DOSSIER

SIMUPHARM.IO — PHASE 4 EDUCATIONAL PILOT & JCE READINESS REPORT

Usability Evaluation Protocol, Diagnostic Instruments & Pre-Specified Statistical Analysis Plan

TARGET JOURNAL	ARTICLE CATEGORY	CODEBASE STATE	DATA INTEGRITY
Journal of Chemical Education	Technology Report / Tip	FROZEN (Zero Code Edits)	ZERO FABRICATION CONFIRMED

📘 **PHASE 4 STRATEGIC PIVOT**

Following the Phase 3.5 Hostile Audit, which broke claims of general predictive accuracy and industrial digital-twin fidelity, Phase 4 establishes the definitive educational evidence package for SimuPharm. The study positions SimuPharm as an **interactive, browser-based chromatography teaching tool** that enables undergraduate pharmacy and chemistry students to safely practice diagnosing classic instrument malfunctions (air bubbles, leaks, voids, contamination).

1. CURRENT SCIENTIFIC CLAIM BOUNDARY

To guarantee intellectual honesty and protect against referee rejection, publication claims are categorized into three rigid tiers:

Claim Tier	Permissibility Status	Permitted Manuscript Statements & Prohibitions
Tier 1: Safe Claims	PERMITTED	• "SimuPharm is a free, zero-install, browser-based educational simulator." • "The platform allows interactive observation of qualitative chromatographic fault symptoms." • "Undergraduate students operated the simulator without extensive technical training, rating it as usable (SUS)." • "Participants demonstrated short-term diagnostic gains on parallel troubleshooting assessments."
Tier 2: Stronger Claims	REQUIRES MULTI-COHORT RCT	• "SimuPharm produces long-term retention of chromatographic troubleshooting skills." • "SimuPharm is demonstrably superior to traditional static textbook troubleshooting manuals." • "Simulation directly reduces instrument error rates and column fouling in physical laboratories."
Tier 3: Prohibited Claims	STRICTLY BARRED	❌ "Industrial Digital Twin" / "Digital Twin" ❌ "Universal predictive accuracy" or "2.07% general HPLC accuracy" ❌ "Artificial Intelligence" / "Machine Learning" / "Bayesian Knowledge Tracing" ❌ "Experimentally validated column backpressure dynamics" ❌ "Mechanistic computational fluid dynamics of pump cavitation or bed voiding" ❌ "Validated library of 300+ pharmaceutical APIs"

2. EDUCATIONAL RESEARCH QUESTIONS & DESIGN

Primary Research Question: *"Can an interactive, browser-based HPLC simulation environment support undergraduate pharmacy and chemistry learners in identifying, interpreting, and troubleshooting common chromatographic system faults?"*

Study Design: Single-group pre-test / post-test exploratory educational pilot ($N = 20–30$ undergraduate students). The study is integrated into a single 50-minute scheduled laboratory period.

Session Phase	Time	Instructional Activity	Instrument / Tool Used
Phase 1: Baseline	00–10 min	Ethical briefing, voluntary informed consent, and 10-item baseline pre-test.	Paper Questionnaire (10-item Pre-Test)
Phase 2: Orientation	10–18 min	Platform orientation; independent student execution of nominal baseline run.	SimuPharm (50% B, $1.0 ext{ mL/min}$, C18 baseline run)
Phase 3: Fault Practice	18–30 min	Interactive exploration of 4 faults: air bubble, leak, column void, contamination.	SimuPharm Fault Injection Console
Phase 4: Diagnostic Task	30–40 min	Practical diagnostic performance task: 4 unknown simulated runs.	Scenarios Alpha, Beta, Gamma, Delta (Timing recorded)
Phase 5: Evaluation	40–50 min	Parallel post-test, 10-item System Usability Scale (SUS), and qualitative feedback.	Post-Test, Brooke SUS Instrument, Qualitative Form

3. DIAGNOSTIC FAULT SCENARIOS (FROZEN-CODE MAPPING)

SimuPharm implements four discrete diagnostic faults in `CONFIG.faults`, mapped directly to John Dolan's *LCGC* troubleshooting literature:

Fault Mode	Code Implementation	Observable Symptoms	Physical Mechanism	Corrective Action
Air Bubble in Pump	`pressureEffect: -20 noiseMultiplier: 5`	Cyclical pressure dips (-20 bar); optical baseline noise spikes; retention jitter.	Gas compressibility dampens piston stroke, causing transient delivery loss.	Open purge valve; prime pump at $5 ext{ mL/min}$ with degassed solvent.
System Fitting Leak	`pressureEffect: -50 rtShift: +0.30`	Steady backpressure drop (-50 bar); $+30\%$ retention time elongation.	Pre-column fluid loss reduces linear flow velocity through packed bed.	Tighten fluidic fittings at pump/injector/column; replace worn ferrule.
Column Inlet Bed Void	`pressureEffect: +30 tailingFactor: 2.50`	Split doublet peaks across all analytes; severe tailing ($T_f = 2.50$); N drop $>50\%$.	Packed bed collapse at column head creates split flow paths for solute band.	Replace analytical column; install guard column; check frit.
Solvent Contamination	`ghostPeaks: true noiseMultiplier: 3`	Spurious ghost peaks in blank runs; elevated baseline drift.	Impurities in aqueous buffer or weak solvent concentrate and elute.	Discard contaminated mobile phase; clean lines; prepare fresh HPLC grade.

4. PSYCHOMETRIC ASSESSMENT & USABILITY INSTRUMENTS

- **10-Item Isomorphic Pre/Post-Tests:** Parallel items mapped onto identical Bloom cognitive levels covering backpressure anomalies, retention drift, peak splitting, ghost peaks, blockage isolation, pH sensitivity, silanol tailing, degassing sequence, void differentiation, and leak localization.
- **4 Unknown Practical Diagnostic Tasks:** Real-time assessment measuring diagnostic accuracy (0/1), latency to diagnosis (seconds), diagnostic confidence (Likert 1–5), physical reasoning, and corrective action.
- **System Usability Scale (SUS):** Standard Brooke (1996) 10-item instrument scored on a 0–100 scale, evaluated against Bangor et al. (2008) empirical thresholds.

5. PRE-SPECIFIED STATISTICAL ANALYSIS PIPELINE

All statistical procedures were pre-specified in `statistical_analysis_plan.md` and verified via Python scripts on labeled demonstration data:

- **Normality Rule:** Shapiro-Wilk test on paired score differences ($\Delta ext{Score}$). If $p \ge 0.05 \implies$ Paired Student's t-test + Cohen's d_z; If $p < 0.05 \implies$ Wilcoxon signed-rank test + rank-biserial correlation r. Both tests reported side-by-side.
- **Normalized Gain:** Hake's $\langle g angle = (ext{Post} - ext{Pre}) / (10 - ext{Pre})$.
- **Non-Fabrication Assurance:** The primary template `participant_data_template.csv` is strictly empty. No student data has been manufactured. Placeholders are preserved in the provisional abstract and manuscript.

6. MANUSCRIPT READINESS EVALUATION (10 GATES)

Readiness Gate	Standard Required	Current Gate Status	Action Required Prior to Submission
1. Software Description	Clear client-side architecture	READY	Complete in manuscript outline.
2. Model Description	Equations & boundary limits	READY	Limits clearly articulated.
3. Prior Art Delineation	Distinct from Boswell (2013)	READY	Positioned around fault injection.
4. Educational Methods	Reproducible 50-min protocol	READY	Instructor guide archived in SI.
5. Assessment Tools	Pre/post tests, rubrics, keys	READY	Archived in Folder 03.
6. Statistical Plan	Pre-registered Python scripts	READY	Verified in Folder 06.
7. Ethics Compliance	IRB exemption checklist	READY	Exemption criteria documented.
8. Participant Data	Zero fake data; blank template	READY	Awaiting real student execution.
9. Reproducibility	Open science TOP compliance	READY	Checklist completed in Folder 08.
10. JCE Submission	Full package readiness	CONDITIONALLY READY	**Execute student pilot to populate placeholders.**

7. MANDATORY FINAL DECISION

CURRENT PAPER THESIS:
SimuPharm is a free, zero-install, browser-based educational HPLC simulator that provides a usable, engaging, and instructionally viable virtual laboratory platform for teaching undergraduate pharmacy and chemistry students how to recognize, interpret, and troubleshoot common chromatographic system faults.

ONE-SENTENCE CONTRIBUTION:
SimuPharm fills a critical gap in pharmaceutical chromatography education by introducing interactive, real-time phenomenological fault injection (pump cavitation, leaks, bed voiding, and contamination) into an open-access web environment, enabling low-stakes deliberate practice in instrument diagnostics.

PRIMARY EDUCATIONAL RESEARCH QUESTION:
Can an interactive, browser-based HPLC simulation environment support undergraduate pharmacy and chemistry learners in identifying, interpreting, and troubleshooting common chromatographic system faults?

STUDY DESIGN:
Single-group pre-test / post-test exploratory educational pilot embedded within a standardized 50-minute undergraduate laboratory session, incorporating 10-item isomorphic assessments, 4 practical unknown diagnostic tasks, and the System Usability Scale (SUS).

TARGET SAMPLE:
N = 20–30 undergraduate B.Pharm (Years 3–4) or B.Sc. Chemistry (Years 2–3) students enrolled in an instrumental analysis or pharmaceutical analysis laboratory course.

PRIMARY ENDPOINT:
Change in troubleshooting assessment score (Delta Score = Post-Test Score - Pre-Test Score, scale 0–10 points) evaluated via paired Student's t-test or Wilcoxon signed-rank test.

SECONDARY ENDPOINTS:
1. System Usability Scale (SUS) continuous score (scale 0–100) and Bangor adjective rating.
2. Practical diagnostic task accuracy (percentage correct across 4 unknown fault scenarios).
3. Median time-to-diagnosis latency (seconds).
4. Diagnostic confidence-accuracy calibration.
5. Qualitative usability barriers and student misconceptions.

CURRENT SCIENTIFIC CLAIMS THAT ARE SAFE:
1. SimuPharm is an accessible, zero-install web simulator suitable for undergraduate chromatography education.
2. The platform interactively displays qualitative hydraulic and chromatographic symptoms of classic instrument malfunctions.
3. The platform achieves acceptable usability among undergraduate student users without extensive orientation.
4. The simulator supports short-term improvements in conceptual troubleshooting scores.

CLAIMS THAT MUST REMAIN PROHIBITED:
1. "Industrial Digital Twin" / "Digital Twin"
2. "Artificial Intelligence" / "Machine Learning" / "Bayesian Knowledge Tracing"
3. "Universal predictive accuracy" or "2.07% general HPLC accuracy"
4. "Experimentally validated column backpressure dynamics"
5. "Mechanistic computational fluid dynamics of pump cavitation or bed voiding"
6. "Validated library of 300+ pharmaceutical APIs"

PHYSICAL HPLC REQUIRED FOR THIS PAPER?:
NO. (JCE Technology Reports do not require physical experimental runs when literature benchmarking is sound and educational utility is demonstrated).

EDUCATIONAL PILOT REQUIRED?:
YES. (Mandatory: Student participant data must be collected to populate manuscript placeholders).

ETHICS APPROVAL REQUIRED/PENDING?:
EXEMPTION AUTHORIZED / PENDING FACULTY CONFIRMATION under standard educational research exemption criteria (45 CFR 46.104(d)(1)).

CURRENT MANUSCRIPT READINESS:
MANUSCRIPT READY SUBJECT TO PILOT POPULATION. (All architecture, protocols, instruments, and analysis pipelines are complete; ready to insert real participant data).

BIGGEST REMAINING RISK:
Failure to recruit an adequate student sample (N < 15) or low student compliance resulting in incomplete pre/post-test pairs during the pilot lab session.

NEXT ACTION:
Obtain formal departmental / institutional authorization, schedule the 50-minute laboratory session with the enrolled student cohort, execute the standardized protocol, and enter audited participant data into the template.

Phase 4 Educational Evaluation Team • Certified October 4, 2026 • SimuPharm.io Publication Due Diligence Panel